

Polynomial Regression Interview Questions & Answers

For Data Scientist, Junior ML Engineer, and Data Analyst interviews.

Q1. What is Polynomial Regression?

Polynomial Regression is an extension of Linear Regression that models nonlinear relationships by adding polynomial features such as x^2 , x^3 , etc. It is still considered a linear model because it is linear in its coefficients.

Q2. Why is it called Linear Regression if it uses polynomial terms?

Because the model remains linear with respect to the coefficients (β_0 , β_1 , β_2 , ...), even though the input features are transformed.

Q3. What problem does Polynomial Regression solve?

It captures curved or nonlinear relationships between features and the target when a straight line is not sufficient.

Q4. What is the equation of Polynomial Regression?

$$y = \beta_0 + \beta_1x + \beta_2x^2 + \dots + \beta_nx^n$$

Q5. What is a polynomial feature?

A new feature created by raising an original feature to a power, such as x^2 or x^3 .

Q6. How is Polynomial Regression implemented in scikit-learn?

Using `PolynomialFeatures` to create polynomial terms and then fitting a `LinearRegression` model on the transformed data.

Q7. What is the degree of a polynomial?

The highest exponent of the feature in the model.

Q8. How do you choose the degree?

Using validation data, cross-validation, and comparison of metrics such as RMSE and R^2 .

Q9. What happens if the degree is too low?

The model underfits and cannot capture the true pattern.

Q10. What happens if the degree is too high?

The model may overfit, learning noise rather than general patterns.

Q11. What is overfitting?

When a model performs very well on training data but poorly on unseen data.

Q12. What is underfitting?

When a model is too simple and performs poorly on both training and testing data.

Q13. How do you detect overfitting in Polynomial Regression?

Compare training and testing performance. A large gap often indicates overfitting.

Q14. Should features be scaled?

Scaling is often beneficial, especially for higher-degree polynomials, because feature values can become very large.

Q15. What evaluation metrics are commonly used?

R^2 , MAE, MSE, and RMSE.

Q16. What is R^2 ?

The proportion of variance in the target variable explained by the model.

Q17. Can Polynomial Regression handle multiple features?

Yes. Polynomial terms and interactions can be created from multiple input variables.

Q18. What are interaction terms?

Features created by multiplying variables together, such as $x_1 \times x_2$.

Q19. What is multicollinearity in Polynomial Regression?

High correlation among polynomial features, which can make coefficients unstable.

Q20. How can overfitting be reduced?

Use a lower degree, regularization, cross-validation, or more training data.

Q21. What is the difference between Linear and Polynomial Regression?

Linear Regression fits a straight line, while Polynomial Regression fits a curve through polynomial feature transformation.

Q22. What is bias-variance tradeoff in Polynomial Regression?

Low-degree models usually have high bias and low variance; high-degree models usually have low bias and high variance.

Q23. What role does train-test split play?

It helps estimate how well the model generalizes to unseen data.

Q24. Can Polynomial Regression model any nonlinear relationship?

Not always. It works well for smooth curves but may struggle with very complex patterns.

Q25. When would you use Polynomial Regression in real projects?

When visual inspection or domain knowledge suggests a curved relationship between variables.

Q26. Give examples of applications.

Salary prediction, demand forecasting, population growth analysis, manufacturing optimization, and energy consumption modeling.

Q27. What are advantages of Polynomial Regression?

Simple, interpretable, flexible, and easy to implement using linear regression tools.

Q28. What are disadvantages of Polynomial Regression?

Overfitting risk, poor extrapolation, increased complexity, and multicollinearity.

Q29. How do you compare different polynomial degrees?

Train models with multiple degrees and compare validation metrics and plots.

Q30. What interview-ready summary would you give?

Polynomial Regression transforms features into polynomial terms and applies Linear Regression to model nonlinear relationships while remaining linear in its coefficients.